

The Light That Entangles: a multi-player exploration of human and post-human entanglement.

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Abstract

Diffraction Analysis is the process of reading phenomena through two contrasting viewpoints in order to create a rounded position based on resonance and dissonance. This paper shares the application of diffractive processes to design methodology through the analysis of a novel three person entangled instrument called The Light That Entangles. Tangible Embodied Interaction and Participatory Sense-making are used as the basis for a human centred exploration of Entanglement, whilst post-human explorations of apparatus form the human de-centering counter-point. The design journey is discussed and initial observations are presented before diffracting the entangled experience through both lenses.

Keywords

Diffraction, Entanglement, Enactive, Participatory, Sense-making, Post-human, Collaborative

1 introduction

There has been some academic discussion around successive ‘waves’ of HCI [9]], this paper explores the exciting intersections where new and seemingly conflicting ways of seeing the world overlap and interfere as they reform each other. One of the issues of navigating this juncture is that whilst applying different theoretical lenses for observing a phenomena might reveal apparently conflicting outcomes, it does not mean that one set of those observations is more valid than the other. The researcher working at the interstices of different approaches needs a specific set of tools to hold and layer the conflicting observations in order to develop useful new knowledge. Diffraction Practice [8],[15]), based around Diffraction Analysis [2] is a method that achieves this by reading phenomena through two contrasting frameworks to seek areas of resonance or dissonance. The Light That Entangles (TLTE)¹ is an instrument based digital music installation for up to three participants. It is one of a series of works exploring how notions of entanglement can be used to create collaborative music making experiences where social engagement resonates as much as as the music produced by the players [16]. TLTE is presented here as an exploration of the intersection between differing design approaches to entangling interfaces. Previous research [18] has shown that Karen Barad’s exploration of post-human entanglement (ibid Barad) can be successfully applied to this field to create specific participant experiences that differ from existing collaborative or co-operative music making systems. There are, however, alternative approaches to entanglement, Palle Dahlstedt explores systems where an action at one point of time binds a musician to specific actions in the future²,

¹I suggest reading this ‘tilt’ for ease.

²Writing is sparse, however see <https://www.youtube.com/watch?v=1g0xs71Sdps> for video documentation.



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whilst for Alva Noë an entanglement is a human process for creating understanding of the world that is intricately bound to the creation of Art [14] . This paper explores how Noë’s understanding of entanglement can be seen to resonate strongly with theories of tangible and embodied interaction that build from a position of enactive cognition [1]; Hanne De Jaegher et al’s [6] consideration of an enactive account for social cognition is used to build on this to argue for a humanist approach to multi-player entanglement. TLTE implements both human and post-human design approaches; audio-modulated light is used to create an embodied experience with strong resonances between individual action and musical effect, whilst a specific sensor assembly within the interfaces creates a common light-space-audio loop that sits at the heart of the systems post-human entanglement.

2 Light Apparatus

The ability to turn light level into an electronic signal lies at the heart of using light as a tool for music making. This has typically been achieved in 3 ways - the light dependant resistor (where the light photons falling on a substrate cause a change in resistance), the photo transistor/diode (a variation on the principle but in semi-conductor form and having different response to the resistive LDR) and the solar cell (where photons are directly turned into electricity). The differences between these array around size and speed of response, range in light level detected and what electronics is required to facilitate its use. A digital camera for example embeds several photodiode sensors into a complex piece of electronic equipment.

The following section gives an overview of the context of several light based NIME’s and it is far from a definitive list and is intended as a source of jumping off points for the reader’s interests. Several projects by Owl Project, a UK based 3 person sound art and sculpture collective, are included as they are integral to the author’s historical practice and work well as scaffolding for the document.

2.1 Light as interface

Some systems use the level of received light to directly control electronics or sound synthesis. The iLog Signal for example (see far right in figure 1 uses LDRs to control a number of interconnected oscillating hex inverter chips³. The addition of switches allows for mode changes by swapping the capacitor used in the circuit, or allowing oscillators to affect each other. Such instruments tend to have constraints designed into them [11] with the deliberate intention of creating a limited instrument that only does a dedicated sonic task. This is emphasised by the circuit that drives the oscillation uses the LDR’s resistance directly resulting in a monophonic system. This can be avoided by decoupling the LDR from a separate synthesis system by using some form of voltage measurement electronics.

Artists have shifted this sound making process from electronics to software, looking to achieve polyphony or more complex control options. Here the voltage change caused by light shining in the LDR or Phototransistor is turned in to a numerical value

³Highlighted by Collins #cite:collins-handmade)



Figure 1: Owl Project iLog range (image courtesy of Owl Project)

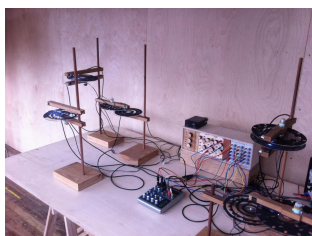


Figure 2: Owl Project Optikit, (image courtesy of Owl Project)

(though analog to digital conversion) and that number used in digital synthesis. Such systems can be embedded into objects using a range of microcontroller technology, “Special thing” for example uses a loop of audio embedded into a Teensy. Leaf Cutter John’s light interface⁴ takes a laptop orientated approach with by scanning multiple light sensors and the resulting values passed to MAX/MSP (via USB) for use in a range of complex synthesis systems.

The Light Organ [5] is another example of a system that uses light levels to control synthesis, but this time adds a time base through the use of slide-able physical scores cut into cardboard. A camera to detect the light levels and software to convert the subsequent image into usable numerical values and so control synthesis. Notably Owl Project’s Optikit⁵ also uses physical scores as a control system, though in the form of spinning disks (2 to form sequences and amplitude control for a modular Eurorack system. These examples show how a looping sequence can be implemented using physical additions to the interface.

2.2 Modulated Light to Sound

The Optikit implements, what could be seen as, a more traditional approach to light-audio-modulation: the analog conversion of changes in light level to sound. This is notably seen in the audio track of cinema and Super8 film, where the audio signal is represented by exposed film on a dedicate section of the negative. The audio is then recreated by shining a light through the moving exposed film and using a photo-diode and amplifier to render the light levels into an audio signal. The Optikit uses spinning disks with the audio represented either as a degree of shadow that relates to the volume of the audio at that moment over the sensor or a block of shadow based on pulse width modulation (PWM). Similar techniques are used by Derek Holzer in his Tonewheels project.⁶

Pulse Width Modulation (PWM) is a way of representing an analog signal in digital form using different length pulses over a fixed time base, the electronics that receives the pulses turns them back into an analog representation, examples being the analogue

⁴weblink

⁵<https://owlproject.com/optikit>

⁶<https://macumbista.net/?tag=tonewheels> and https://macumbista.net/files/derek_holzer-tonewheels.pdf



Figure 3: Owl Project Logpile

voltage output of an Arduino. Motor control through AC and DC signals and even Audio Output from microcontrollers can all use PWM signals based on an output being ON or OFF for a period of time. Modern LED based bike lights use this principle to create different brightnesses in a way that maximises battery power. The fact that this can be turned into sound has been exploited by several artists; such as the Owl Project iLog Photosynthesisor performances⁷ that used a photo cell connected to a mixer and a range of objects such as bike lights and LED fans to create a sonic tapestry or beats and drones. In modulated light to sound systems the light literally transmits audio in a data form (as intensity or encoded PWM, say) through physical space. This allows for the performer to easily combine two light-audio sources by pointing them at the same sensor and to also control the mixing effect by angling the source or moving it further away from the sensor.

Light modulation need not happen at source but by thinking of the sensors as a ‘light microphone’ light can be sensed after interacting with reflective (e.g. a cymbal) or translucent (e.g turbid water) materials to reveal sonic interpretations of the objects movement or of particles within the water.

2.3 Modulated Light Shared Apparatus

Driving an LED directly from an audio source such as a mp3 player or laptop, requires amplifying electronics, however this is not required if the audio is being generated in a digital PWM form, from say a simple microcontroller such as an Arduino Nano using, say, the Arduino Mozzi library⁸. PWM signals can easily be turned into analog ones using a low pass filter, however the microcontroller output is also suitable for directly driving an LED. The resulting output is brighter, and so providing a greater transmission range, than using the analog voltage level of the audio.

The Owl Project Logpile⁹ uses this technology to create a co-operative musical experience based around the concept of sitting around a camp fire. This work is literally made up of a pile of logs (see image #ref:image-op-logpile) each studded with super bright white LED’s, with an Arduino NANO based synthesiser embedded inside. Each log represents a different sound source that vary from filtered white noise to tones (different related pitches along a log) to slower beats. The physical construction of Logpile causes the LEDs to create overlapping areas of audio-transmitting-light which can be picked up, and rendered audible, using the previously mentioned iLog Photosynthesisor. Players are free to explore the Logpile soundscape by moving the sensor log above the Logpile, creating their own mix and composition as they explore.

The installation has 3 sensor iLogs each of which is connected to a dedicated wooden speaker horn, so individuals can easily

⁷<https://www.youtube.com/watch?v=3MBnnZnGak8>

⁸<https://sensorium.github.io/Mozzi/>

⁹<https://owlproject.com/logpile2015>

perceive what sounds are their responsibility. They can explore the audio further (and individualise their own voice) using the sensor log's in built resonant filter using the large wooden knob on the iLog's interface. The atmosphere of sitting around a shared campfire is emphasised through the strategic placement of the sensing log cables and small stools to sit or rest the iLog on after use.

2.4 Implications for performance and installation

Light based instruments, such as those discussed, resonate strongly with aspects of Armstrongs' description of Tangible and Embodied Interaction [1]; this can be revealed by reflecting on video documentation, alongside experiences as an audience member and performer of related work. Audiences can create correlations between seeing a light shone at an interface and hearing a resulting audio response, much the same way as they can associate between a violinist's movement and music without understanding the physics of the string's vibration. A light instrument performer can also use several strategies to demonstrate the interface to the audience, such as 'testing;' or 'tuning' at the start, possibly as seen in other novel instruments for musical expression. It can be argued that this resonance between previous and current experience also underlies part of the audience experience for the mechanically modulated light systems such as Derek Holzer's and Owl Project's spinning disks. Some artists make their processes more tangible for the audience, participants through workshops or drop in play session. Here participants are allowed a direct exploration of the means of light modulation and its effect on the audio produced by the system through cutting shapes and hearing the resulting effect on the audio. This functional learning could lead to an enhanced inactive audience experience.

The Logpile however does not have any such obvious control mechanism, some of the embedded LEDS flicker but other than that the means of audio production is kept mysterious by the artist. This, I argue, shifts the experience from one of 'learning about function' to one of 'learning about being' [10],[12]. Both modes can be exploratory processes and both can build on and transcend a participant's previous experiences, however the differences between focusing on the means of production and physically exploring the production of un-knowable light-sound space around the Logpile are not the same cognitive process. Here I ascribe functional learning with direct mechanical or sensory effect, and learning about being exploring more affective dynamics.

This difference has a major effect when considering the multi-user experience of a work such as the Logpile alongside a shared function-learning or composition task, using a shared sequencer interface system such as the Daisyphone [4]. Here there is an external task or system that the participants engage with alongside each other rather than acting directly upon each others' experiences. We can use Armstrong's model for tangible and embodied interaction, to explore this further (figure 4 version 1), specifically by adapting Maturana and Varela's model of sensory-motor control for enactive cognition [13], that sits at the heart of it.

The separation between 'agent' and 'world' is key to the enactive model of cognition. Agents ensure their survival through adapting their connection to the world through a motor sensory control loop. If we consider two people co-operating at an interface, then that interface can be considered as the 'world' that both

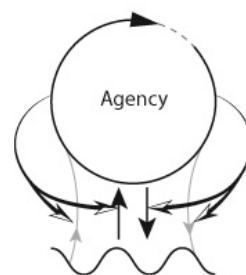


Figure 4: Enactive Sensory-motor Control, reproduced from The Tree of Life, Maturana and Varela

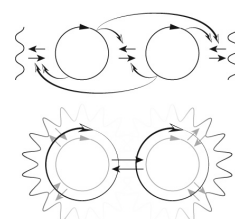


Figure 5: Enactive model for Participatory Sense-making, reproduced from de Jaegher et al [6]

participants are engaging with and the image redrawn as in version 2, play alongside #ref:image-playalongsideside). This viewpoint places the shared world interface at the centre of the engagement, where it acts as a conduit for mediating co-operative work. Shared musical interfaces designed from this perspective can facilitate a range of interesting and often useful mutual engagements [3], such as working near each other (where spatial design metaphors allow), or mirroring or directly editing each other's work. Such mutual actions facilitate both positive and negative user experiences. Mutual free editing, for example, allows for any team member to freely change group work, but without a group validation process this can lead to frustration when one person overrides the original group members actions [3]. Spatial interface metaphors encourage opportunities for virtual approaching someone and mirroring or demonstrating adapted ideas, but they also carry territorial issues [7]. These two examples suggest that something very complex is occurring at these junctures: areas where the agents acting on the shared world are so congruent that there appears to be little division between the agents' individual coupling. The agents are effectively acting on each other.

De Jaegher et al [6] greatly expand upon this approach to develop an enactive account for social cognition; where a key premise being that social agents are also able to act directly upon their own and others' coupling to the world (see figure 5). This mutual action in turn feeds into both the agents' motor-sensory control loops providing many more opportunities for feedback, adjustment and development. The caveat being, for anyone following the suggestion that more routes for mutual engagements and participant interconnections automatically leads to more co-operation, is that increased opportunities for something does not necessarily mean more of that something being produced. In fact as we have seen from the mutual engagement research, the opposite might in fact occur. Imagine a small tabletop with

a grid of 25 sensors that has been configured to act as a musical interface. This table has a number of cushions set around it inviting multiple people to play together. On one hand we could think of this tabletop as providing multiple routes for participatory sense-making and musical collaboration. However, I argue, that each of the 25 sensor controller represents its own enactive world within which players may or may not decide to co-participate, potentially choosing to explore a different area of the tabletop altogether. The designer wishing to encourage mutual participation naturally explores ways of connecting and entwining these sub-worlds though a group task, or connecting the sensors to shared synthesis controls for example. I suggest that even then the individual controls represent individual enactive worlds, even if they operate different parts of the same sonic machinery. Adding multiple controls enables multiple people to play alongside each other, but it also enables the potential emergence of personal roles and individual spaces.

There is no doubt that the enactive approach proves useful insight into the relationship between shared system interface design and the way participants can mutually form understanding of the system function whilst engaging with it. Shared experiences can emerge from highly complex multiple interconnections; each of which representing an individual enactive space within a system. Additionally it is clear that something dynamic happens when more than one person engages with each others' enactive processes to learn about being with each other within the world rather than controlling it. This suggests that we can create new forms of shared world by reducing the potential interconnections between players, but at the same time making these shared spaces richer with many degrees of freedom for the players to explore. It is in configuring these highly mutually affecting experiences that system designs that explore different views on entanglement can prove useful.

3 The Light that Entangles

The Light That Entangles (TLTE)¹⁰ is a digital musical instrument for up to three players. The work is part of a practice research process into multiplayer instruments called Entangled Instruments [16] that seeks to apply post-human theories of entanglement to instrument design. This section presents the technical background to TLTE and a reflection on the design processes (both enactive and post-human) that lead to the final outcome, the section concludes by diffracting these design processes.

3.1 Technical Description

The Light That Entangles (TLTE) is an instrument based installation with three interfaces that complete an entangled apparatus through player action. The potential for entanglement is designed into the hardware of the interfaces themselves: however the nature of this changes when one person engages and, it shifts again with two participants. When three players are within the apparatus, then the individuality of the three separate elements becomes very hard to distinguish. Individuals need to work to separate their own functionality from the group experience of being together.

The TLTE interface is a laser cut 26 sided figure called a rhombicuboctahedron, made up of 18 square and 8 triangular sides (see 6, the LEDs that stud the object broadcast simple tones as audio-encoded light (as discussed previously), with each face

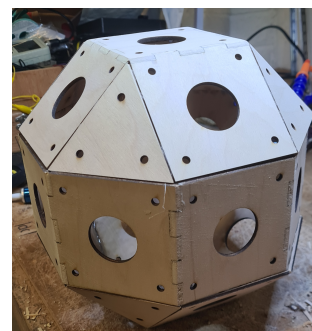


Figure 6: Full TLTE interface in construction.

broadcasting a different frequency.¹¹ One triangle also has a power connector, this acts as a reference point for the players and as recharge connection for the object. Internal sensing ensures that a module only becomes active whilst held, ensuring that is difficult for a single player to experiment with more than one interface. At twenty-three cm in diameter the interface is designed to be held in to two hands and offered to co-players. The square sides also have solar cells which act as receivers for audio modulated light, all 18 of which are mixed equally to create a single mono audio signal, this is then broadcast to a central audio mixer. This mixer combines the signals from all three interfaces equally and then outputs the resulting mono signal to either 4 speakers, to surround the players in a directionless audio field, or 3 wireless headphones (one for each player).

3.2 Design Reflection

A deliberate decision was made in TLTE to not measure and map relative positions of the interface objects however the journey to this conclusion was not direct. A previous Entangled Instrument (Elastiphone [17]) used tracking a single hand from each of two players and then mapping the relative angle and separation, alongside a 3d position of the mid point between the hands, to synthesis. Early sketches for TLTE show an attempt to recreate the information from the earlier hand-tracking systems using embedded 9 degree of freedom IMU and Bluetooth signal strength to calculate relative and positional angular data between objects. The aim being to gather as much information about what individual players were doing with the object they had and then to entangle them mathematically or systematically through mapping. Issues with using Bluetooth range-finding being noted experiments using light or IR or other forms of positional sensing were considered.

However, observations from the Elastiphone studies highlighted a link between players' awareness of each others' hands and the effect that the distance and angles between the players hands had on synthesis control. The players were often observing each others' hands and often commented that the experience was 'play fighting'. This heightened awareness of a specific part of the body seems to have been at the expense of awareness of the co-player as a person. The inter-player relationship appeared based on a functional learning of the enactive world that the measured distances and angles represented.

The move away from these mapping strategies was based on Karen Barad's reflection that the means of observation affecting potential outcomes of a system is applicable to within the

¹⁰Reading or pronouncing this as 'tilt' is helpful.

¹¹Initial designs explore a Pentatonic scale with a Pythagorean Temperament.

social context as well as the realm of quantum mechanics [2]. This resonates with another Entangled Instrument: the Perceptron [18], where absolute hand positions and machine learning entangles the players. The change from relative to absolute measurements avoids creating an enactive world based on acting on and between each other as fixed points, by creating one that the players are acting within. It maybe that players interpret certain co-ordinated movements between themselves as being mapped to a specific feature in the soundscape, but this has to been pre-conceived and programmed by the designer.

With the decision made to deliberately not detect the position, location or angle of the interface objects and then use that in a separate synthesis system, the interfaces were then redrawn as areas of affect that interplay with each other rather than discrete objects. This directed the design process to exploring the potential for inter and intra-action as facilitated by the interfaces' skin and the space between, without any knowledge of the objects' relative or absolute position in space. This change effectively shifts the interfaces from being focal points (and so the participants) and thus a technology need be sought that, literally, creates a mutual network of forces with each able to act on the whole apparatus directly.

3.3 Audio modulated Light Entanglements

The solar panels in the final interfaces are integral to creating the shared space for engagement in TLTE. Each square face has a panel and all the panels in an interface are mixed equally before the resulting audio is transmitted to a central point. Unlike in the Logpile where each interface has its own speaker, here all the three of the objects' signals are equally mixed and the resulting single channel of audio is broadcast to the participants. This closing of the system-audio-participant loop can be achieved either through wireless headphones or through 4 speakers surrounding the intended play space. Both these options stem from experience with previous Entangled Instruments and the need for the players to feel immersed in the experience. The deliberate non-individualisation of the audio feed creates a single shared apparatus that each participant can intra-act with equally.

#image:image_tlteSystem

Although the potential for acting mutually within the system is created by the common solar-panel-audio bus; it is the modulated light that each interface emits that completes the apparatus of entanglement. Each individual object transmits a range of audio sources from its facets via the LEDs, but this is only fully heard the modulated light falls on a different objects' solar panels;¹² this second object is also emitting light and so it too is constantly acting within the shared apparatus. The system diagram in figure #ref:image_tlte_system illustrates the whole apparatus. From this we can see that by manifesting the entanglement deeply in the physical world, what is essentially a single connection between players can facilitate the rich and complex range of freedom discussed earlier. Although an individual participant rotating their interface will always create, a motor-sensory loop closing, shift in the audio. This change is dependant on a resolution of all the forces that the modulated light emitting from each LED cluster represents. The individual therefore never acts completely alone but always within the overall apparatus of the entanglement.

¹²It is just about possible to reflect the light from an interface back onto its own solar panels. We could think of the players' entanglement with their own interface as part of the whole apparatus.

Early reflections from playing TLTE in a development mode with myself, or with others and observing others playing are revealing. These reflections are based on studio based tests, public demonstrations and a solo performance that shifted to audience participation. These all involved either two interface 'lids' (5 LED and solar panel facets each) or two lids and a complete TLTE interface. Whilst playing solo: from an tangible embodied perspective there was a strong sense of understandable control based on proximal engagement with the objects, but this is tempered by a sense of it happening within a wider degree of possibility. Moving between tones is smooth as the modulated light spreads out from the LEDs at about 120 degrees and the placement of tones on neighbouring facets is a key directive in melodic creation. Holding the lids in both hands did not create a controller and receiver dynamic, as using a single bike light and solar panel does naturally. As such gestures felt like fluid oscillations rather than authoritative actions. I was very aware of my position of experience, whilst inviting a series of individuals to play and take over control of the interfaces from me. Interesting I did not feel the need to pause and create a specific musical place for my co-players to learn basic instrumental functionality, this was in stark contrast to playing other more directly interactive Entangled Instruments. However there were moments when one person deliberately brought their interface close to another, this had the effect of amplifying one particular set of tones. Such actions could be seen as being to create a specific composition or as constraint making as part of a functional learning process. During dyadic engagements players were observed sharing their attention between their co-players objects and their co-players faces indicating both a connection with the objects and the player. However when all 3 players were engaged there seemed to be more of a tendency to look in a less direct manner and rely on wider perception. The two lids and whole interface arrangement had a strong effect of balance within the system. The two lids being seen as peripherals - one audience player wanted to confirm that playing the lids without the full object was the same as a lid and full interface, only more limited. This dynamic was also seen in the body postures between the three players - with the person holding the full interface more likely to proffer it for the others to orbit around. This emphasises the need for all the interfaces to be the same. The engagement between the two lids tended to be more 'shield' like and led to more play fighting as seen elsewhere. Future iterations will resolve to 3 full interface objects for a three person installation performance, and the 2 lids will form the basis of a solo performance interface.

4 A Diffraction of Participatory Entanglement

If we consider the individual, then it can be argued that they can also ascribe their actions and the subsequence sonic changes to be part of their personal motor sensory control loop, yet in doing so they are also modifying their co-participants' connections to a shared world or an entanglement. Here I argue that we could use both participatory sense making and post-human entanglement as equal elements in TLTE's design ethos. Both lenses reveal equally useful but apparently incompatible ways of considering the design.

Diffraction analysis is the layering together or the reading of an issue through of two contrasting modes of thought [15] with the aim of building a deeper understanding beyond mere differences. Stemming from Karen Barad's post-human theories of agential

realism [2] it is used here to explore the collaborative experiences of TLTE through both humanist and post-human ideas of entanglement.

Previously we presented the technology within TLTE from an enactive perspective, this intrinsically human centred approach places the individual's experience at the centre of the analysis. This is most easily understood through a strong sensory-motor correlation that comes from angling areas of audio-modulated light at a sensor and then hearing the resulting change in the audio. But we have also indicated that the wider system created by the interconnected and equal mixing between all of the solar panels, represents a shared apparatus that the players all exist within all the time. This post-human approach de-centres the individual human from the analysis and it is this entanglement that the experience emerges from.¹³

Notions of entanglement are not limited to the post-human approach and we can see this within TLTE through the participatory sense-making model discussed earlier. Here we recognise the embodied movements of the individual players rotating and moving their interface, but note that TLTE also requires that they bring the interfaces into a mutual configuration where one individuals' light falls upon another interface in order to hear anything. This ensures that the individual modifying their own configuration to the world also re-configures their co-player's coupling to the shared sound world. They are, therefore, entangled in a very direct sense of the word. It is interesting to note that it through 'An Entanglement' such as this that writers such as Alva Noe argue we "construct meaning of the world", in fact Noe takes this further by arguing that this understanding is the basis of the concept of Art and how we re-make ourselves through engaging with an entanglement.

We can expand on De Jaegher's approach to social cognition by viewing participatory sense-making as an interaction where two people influence each other's connection, as seen in TLTE. We can think of the players moving along an "I-We" axis of awareness. They remain individuals but continuously shape and redefine each other through their co-play. The core of this process is how each player directly influences and adapts both their own and their co-player's connection to the shared world. An individual's ability to act at any given moment—and their sense of being part of this shared world—depends on the possibilities created by this connection. At the same time, this connection is shaped by their co-player. This creates a dynamic view on agency that fluctuates and shifts between the players based on their consecutive actions. Which in turn suggests that the collective 'We' experience emerges from the process of negotiating mutual engagement. From this perspective it is the structural forms of the shared world that frame the negotiation, the LED's, the angles of the interface shape and the solar cells that give rise to specific affordances. Attunement between the players then emerges from embodied interaction between the players situated within these constraints.

The post-human approach, which emphasises the agency of entanglement within the process of re-enacting a world, offers a different perspective on TLTE. Here, the focus shifts to the light itself and the shared apparatus that gathers and transforms it, through the LED and solar cell network, into audio. While human actors are required to participate in this cycle of creation and transformation, they do so within the framework it establishes.

¹³Note that post-human theories do not remove or replace humans but accepts that there are more-than-human material and non-material elements.

There is no predetermined action before the apparatus is completed through humans and interfaces engaging. In fact, I argue that the apparatus does not fully exist until this convergence occurs. Since this completion depends on multiple players, the collective experience can be understood as preceding the individual one. This perspective reverses the direction of travel on the "I-We" axis. The resulting re-framing of the experience is emphasised by any gesture with one of the interfaces re-configuring the whole entanglement. The individual experience then requires deliberate effort to isolate from the group (we orientated) experience. Agency is still seen as being dynamic but no longer dependant on one individual's effect upon another but constantly emerging from being in the shared world. Arguably this form of engagement should result in lower fluctuations in individual agency and is simply felt to constantly re-form. Creating interfaces with a diminished focus on personal agency may be key to how a sense of being with ones co-players might emerge.

5 Conclusions and future developments

We seem to have arrived at an interesting position then for TLTE, where the design includes a strong route for, enactive based, functional engagement; however the same could be also be said for considering the participants as exploring within a post-human entanglement of force and influence. These two approaches give rise to apparently contradictory explanations of whether the individual experience emerges from the collective or visa-versa. In reality it is expected that, as seen in previous studies of entangled instruments, participants will oscillate between the two modes of experience. Where the interface design ethos influences how likely either will emerge at anyone time. In the case of TLTE, as both human and more than human approaches are equally present then either one could be likely to be observed. The initial observation seem to bear this out, with both functional or direct interaction and simply being with each other observed. However it is too early to draw conclusions as each player needs to have the same interface for equity of potential engagement. The aim here is to present the underlying approach. The design journey for applying both a tangible embodied and post-human design approach to a multi-player entanglement has been presented here. Starting with the decision to deliberately avoid measurement, from a human perspective, within the interface design raised new challenges to overcome. The urge for the designer to use their own experience to constantly pre-empt what functionality the user might want is powerful. Wishing to be able to create a composition where specific notes are sustained, for example, could lead to adding a 'hold' button to the interface. However this immediately raises questions of empowering one individual over the others. The post-human approach of agency emerging through entanglement suggests different solution, such as a shared reverb unit. To design with such a duality of approach therefore involves holding both methods in constant diffraction. This, obviously, initially requires more effort and risks creating an experience that is confusing and flavourless. On the other hand if the design resonates both a human and more-than-human way then it could facilitate very rich and poetic experiences. Future research can query this further as the finer details of The Light That Entangles are explored, such as changes of tone or the addition of envelopes to create beats or the individualisation of one interface over another. We are sharing the underlying design position here in order to see what other researchers discover as they create their own implementations and diffractions.

6 Acknowledgments

Thanks redacted

7 Ethical Standards

This research has been conducted ... redacted for review

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